HYDRALAZINE

Pregnancy: severe hypertension

Lupus-like syndrome

CHEMICAL NAME 1-hydrazinylphthalazine (hydrochloride)
CLASS Direct-acting arteriolar vasodilator
TRADE NAMES Apresoline · Hydralazine HCl injection

THERAPEUTIC USES / INDICATIONS

- Severe hypertension in pregnancy / pre-eclampsia (IV).
- Heart failure (with isosorbide dinitrate), especially Black patients.
- Resistant hypertension (with β-blocker and diuretic).

ADMINISTRATION

- **IV:** slow injection of 5–10 mg diluted in 10 mL saline, repeat after 20–30 min if needed, or infusion; monitor BP and HR closely.
- **PO** with food (↑ absorption).
- Not as monotherapy long-term (reflex tachycardia, fluid retention).

DOSAGES

- **Severe hypertension in pregnancy:** 5–10 mg IV slowly, repeat after 20–30 min if needed, or infusion 200–300 mcg/min then 50–150 mcg/min.
- **PO hypertension:** 25 mg twice daily, up to 50 mg twice daily.
- **Heart failure (with isosorbide dinitrate, e.g. BiDil 37.5/20 mg):** 1 tablet three times daily, titrated to 2 tablets three times daily (75 mg hydralazine tds).

PHARMACOKINETICS

- **Onset:** IV 5–20 min; PO 20–30 min. **Duration:** 2–6 h.
- Metabolised in liver by N-acetylation (slow acetylators ↑ levels and lupus risk).
- Excreted in urine. **t½:** 2–4 h.

PHARMACODYNAMICS

- Direct relaxation of **arteriolar smooth muscle** → ↓ afterload.
- Reflex sympathetic activation → tachycardia, ↑ renin and fluid retention.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Reflex tachycardia, palpitations, headache, flushing.
- Nausea, dizziness.
- Fluid retention.

ADVERSE EFFECTS (SERIOUS)

- Drug-induced lupus (arthralgia, rash, fever); dose > 100 mg/day, slow acetylators.
- Precipitous hypotension → fetal distress.
- Angina in coronary disease.
- Peripheral neuropathy (pyridoxine deficiency).

CONTRAINDICATIONS & CAUTIONS

- SLE.
- Severe tachycardia; high-output heart failure; cor pulmonale.
- Dissecting aortic aneurysm.
- **Caution:** coronary artery disease, cerebrovascular disease.

KEY INTERACTIONS

- **Other antihypertensives, nitrates:** ↑ hypotension.
- **β-blockers:** used together to counter tachycardia.

NURSING MANAGEMENT

- IV: BP and HR every 5 min until stable, then every 15–30 min.
- Continuous fetal heart monitoring in pregnancy.
- Pre-load with fluid (e.g. 250–500 mL) as per protocol to avoid sudden fall.
- Long-term: monitor ANA, joint pain.
- Watch for headache, palpitations.

PATIENT EDUCATION

- Report joint pain, fever, rash (long-term use).
- Rise slowly.
- Report chest pain or palpitations.

KEY POINTS

- Direct arteriolar vasodilator.
- Reflex tachycardia, headache.
- IV for severe pre-eclampsia hypertension.
- Lupus-like syndrome with long-term high dose.

SOURCES CHECKED **[345]** Hydralazine 20mg Powder for Concentrate for Solution for Injection/Infusion (UK SmPC) **[346]** Hydralazine 25 mg Film-coated Tablets (UK SmPC) **[311]** NICE NG133 **[347]** isosorbide dinitrate and hydralazine hydrochloride, oral (US label) **[348]** BiDil, oral (US label)
Full references p. 432–457